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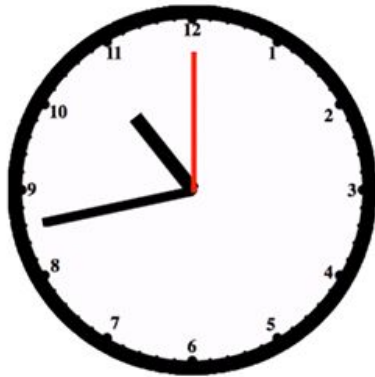
Behaviour Modeling

- Model Object behaviour assuming that it is independent of user's actions
- Large Environments when user cannot control all interactions
- Ex: Office with sliding door, clock, calendar, thermostat, etc



This is one method of Behaviour modeling, using external sensors:

System time:

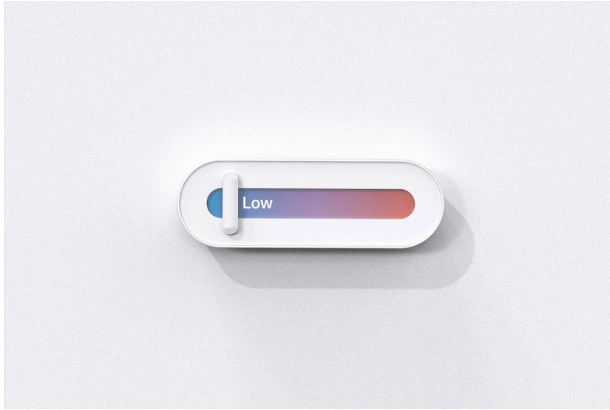


Clock



Calendar

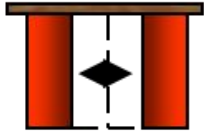
Thermostat



Proximity sensor



Definition An virtual human (or agent) is a 3D character that has a human behavior. Groups of such agents are called crowds and have crowd behavior.



Interactive object



Intelligent agent



Group of agents

The simulation level of autonomy (LOA) is a function of its components

Thalmann et al. (2000) distinguish three levels of autonomy. The simulation components can be either “guided” (lowest), “programmed” (intermediate) and “autonomous (high)

Simulation LOA = f(LOA(Objects),LOA(Agents),LOA(Groups))

guided agent needs a user-specified path in order to travel from one location to another

Example: A guided door has to have its degree of opening controlled by the agent

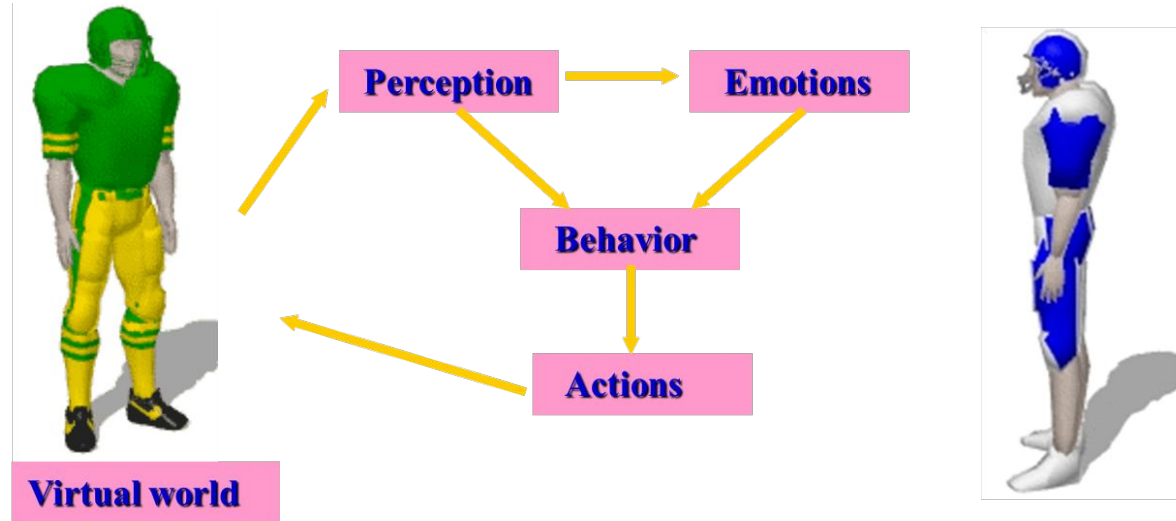


An autonomous agent perceives information about the surrounding virtual environment and decides what path to follow accordingly

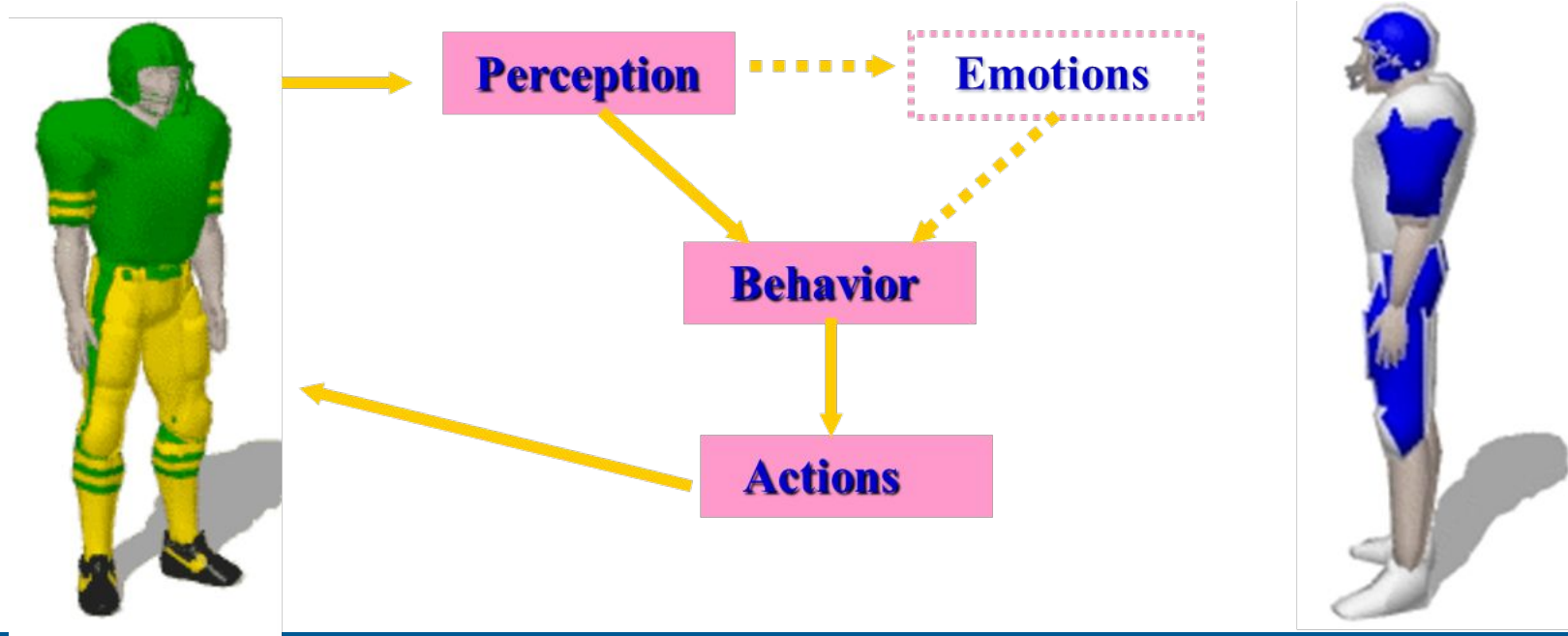
Example: Autonomous agents, such as the virtual football player, need to perceive their environment (in this case the opponent) in order to take appropriate actions

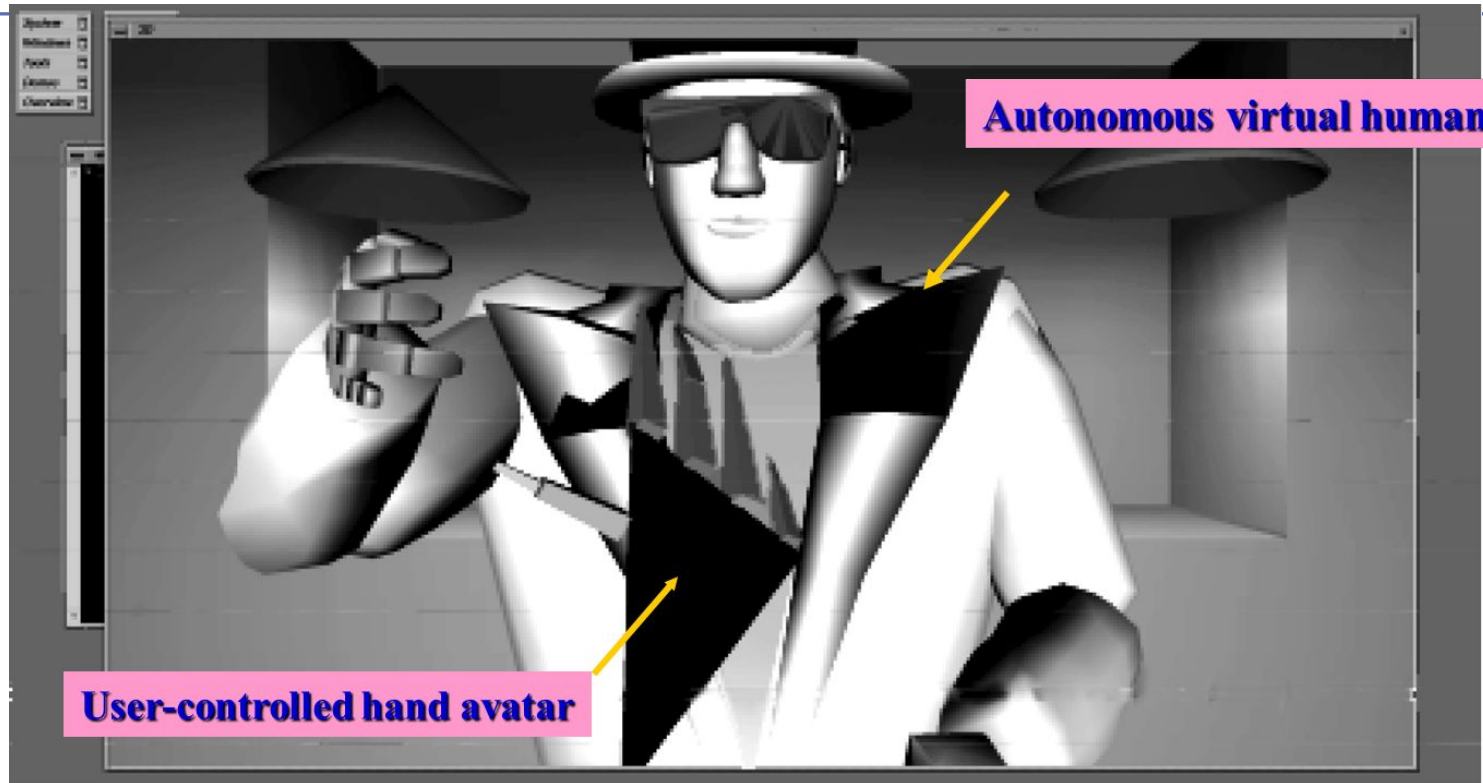


A behavior model composed of perception, emotions, behavior, and actions;
Perception (through virtual sensors) makes the agent aware of his surroundings.



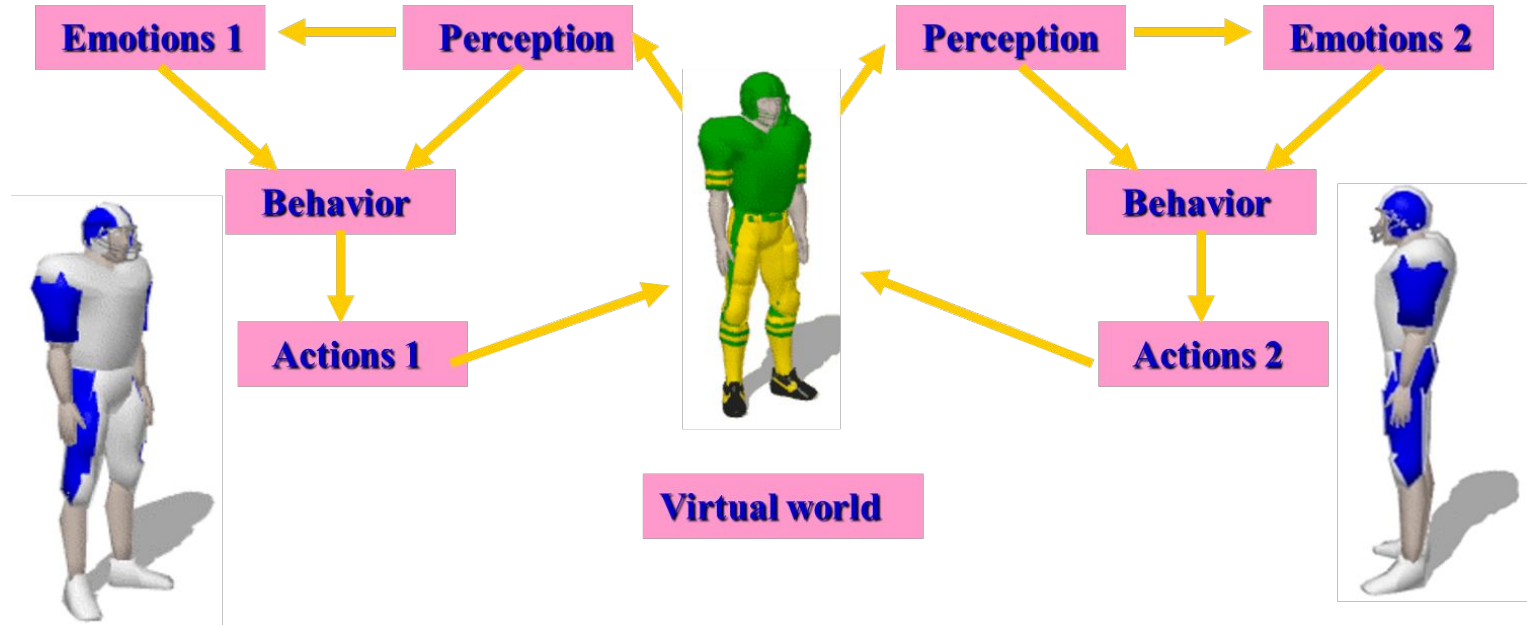
- A direct link between perception and actions (following behavior rules (“cells”));
- It is at a lower level
- Does not involve emotions.
- Example: tackle his opponent every time he sees him





**Another example of reflex behavior – “Dexter” at MIT
[Johnson, 1991]: Hand shake, followed by head turn**

- A subjective strong feeling (anger, fear) following perception;
- Two different agents can have different emotions to the same perception, thus they can have different actions
- Thus at a higher level than simple reflex behavior



- Crowd behavior emphasizes group (rather than individual) actions;
- Crowds can have guided LOA, when their behavior is defined explicitly by the user;
- In this case the crowd has a common goal, but the agent has a way to perceive its immediate surroundings (within the crowd) and react

Political demonstration



- Alternatively, the crowd may be programmed to follow its leader
- Or they can have Autonomous LOA with behaviors specified by rules and other complex methods
- In this case Group perceives info on its environment and decides a path to follow to reach the goal

Building Evacuation



(a) Virtual human modes



(b) The virtual humans travelling along the designated path